

LEEMING, United Kingdom

WMO: 032570

Lat: 54.2969N

Lon: 1.5331W

Elev: 132

StdP: 14.63

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.9	26.9	20.8	15.6	27.7	23.1	17.5	30.4	36.8	49.5	32.4	46.9	3.0	160	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	76.9	64.3	73.3	62.7	70.3	61.1	65.7	73.9	64.1	71.3	62.4	68.6	9.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	85.6	69.0	61.1	80.7	67.3	59.6	76.6	65.5	30.6	73.9	29.3	71.4	28.1	68.6	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.3	23.4	20.7	DB	17.8	82.9	4.0	2.8	14.9	84.9	12.6	86.6	10.3	88.1	7.4	90.2	(4)
(5)			WB	17.2	69.1	4.0	2.1	14.3	70.7	11.9	71.9	9.7	73.1	6.7	74.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.1	39.4	40.2	42.8	46.5	51.7	56.9	60.9	60.3	56.1	50.4	43.8	39.6	(6)	
	DBStd	9.36	5.70	5.78	5.25	5.21	5.17	4.61	4.29	4.28	4.22	5.26	5.80	6.75	(7)	
	HDD50	1591	328	278	229	128	42	3	0	0	3	59	196	325	(8)	
	HDD65	5830	792	696	688	554	413	247	141	156	269	452	635	787	(9)	
	CDD50	1268	1	2	6	24	94	210	338	321	186	72	11	3	(10)	
	CDD65	31	0	0	0	0	0	3	14	12	2	0	0	0	(11)	
	CDH74	247	0	0	0	0	6	33	114	84	9	1	0	0	(12)	
	CDH80	29	0	0	0	0	0	3	12	14	0	0	0	0	(13)	
(14) Wind	WSAvg	9.2	10.7	10.8	10.3	9.2	9.2	8.4	7.8	8.0	8.3	9.0	9.3	9.7	(14)	
(15) Precipitation	PrecAvg	24.7	2.2	1.7	1.7	1.8	1.9	2.2	2.1	2.5	2.1	2.3	2.3	2.2	(15)	
	PrecMax	36.5	4.9	4.3	4.3	5.4	5.2	7.0	5.5	6.5	6.9	5.3	6.0	5.1	(16)	
	PrecMin	17.4	0.3	0.1	0.3	0.1	0.2	0.4	0.4	0.6	0.2	0.6	0.6	0.6	(17)	
	PrecStd	3.3	1.1	0.9	0.9	1.2	1.1	1.5	1.2	1.2	1.4	1.1	1.1	1.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	56.2	60.3	68.1	73.9	78.8	81.7	82.1	75.1	66.4	59.4	56.8	(19)	
		MCWB	51.2	51.7	51.9	56.0	61.3	64.3	65.6	65.6	63.8	60.0	55.5	53.3	(20)	
	2%	DB	52.4	53.3	56.6	62.2	69.2	72.9	77.3	75.6	70.0	62.8	56.9	54.2	(21)	
		MCWB	49.0	49.3	50.1	53.3	58.8	61.7	64.5	64.3	61.1	57.6	53.4	51.0	(22)	
	5%	DB	50.1	51.2	54.0	58.9	65.1	69.5	73.7	71.4	66.6	60.5	54.8	51.9	(23)	
		MCWB	46.9	47.3	48.7	51.3	56.1	59.8	63.0	62.4	59.0	56.2	51.5	48.9	(24)	
	10%	DB	48.1	49.3	51.8	56.1	61.7	66.3	70.2	68.6	64.1	58.6	52.7	49.6	(25)	
		MCWB	45.3	45.6	46.9	49.7	54.0	58.1	61.3	60.6	57.5	54.6	49.8	46.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.7	51.9	53.1	57.2	63.0	67.1	67.9	67.9	65.2	61.3	56.4	53.5	(27)	
		MCDB	54.3	55.3	57.7	64.0	72.1	75.6	77.8	77.2	72.3	65.0	59.1	56.2	(28)	
	2%	WB	49.4	49.8	51.1	54.4	59.8	62.9	65.8	65.4	62.6	58.8	54.0	51.4	(29)	
		MCDB	52.1	53.1	55.6	60.7	67.6	70.2	74.1	73.2	68.5	61.8	56.3	53.8	(30)	
	5%	WB	47.4	47.8	49.2	52.2	57.2	60.8	64.1	63.7	60.3	57.0	52.0	49.3	(31)	
		MCDB	49.8	50.8	53.3	57.6	63.5	67.4	71.7	70.1	64.6	60.0	54.3	51.6	(32)	
	10%	WB	45.4	45.9	47.4	50.3	54.9	59.0	62.3	62.1	58.5	55.1	50.1	47.0	(33)	
		MCDB	47.8	48.9	51.4	55.5	60.5	65.1	68.9	67.1	62.6	58.0	52.3	49.3	(34)	
(35) Mean Daily Temperature Range	MDBR		9.2	11.0	13.2	15.4	15.9	15.4	15.9	15.5	14.7	12.5	10.7	9.8	(35)	
	5% DB	MCDBR	11.0	11.9	16.5	19.9	22.0	21.6	21.9	19.7	18.9	13.5	11.6	11.2	(36)	
		MCWBR	10.1	9.6	11.3	12.2	13.1	12.1	11.1	10.1	11.3	9.6	9.8	10.0	(37)	
	5% WB	MCDBR	10.8	11.5	15.3	18.3	20.3	19.2	19.5	17.7	16.2	12.9	11.0	11.0	(38)	
		MCWBR	10.1	9.8	11.3	12.1	12.9	11.8	10.8	9.9	10.8	10.0	9.8	10.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.322	0.337	0.368	0.395	0.395	0.397	0.397	0.385	0.365	0.347	0.321	0.306	(40)	
	taud		2.305	2.359	2.280	2.228	2.275	2.294	2.299	2.349	2.411	2.425	2.369	2.284	(41)	
	Ebn at Noon		196	237	254	263	270	270	267	264	254	231	197	175	(42)	
	Edh at Noon		18	24	32	39	39	39	38	34	29	23	17	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		193	396	740	1171	1495	1522	1469	1213	877	488	239	148	(44)	
	RadStd		23	46	95	128	86	133	132	79	79	42	24	15	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page